Today's agenda

- No. of iterations
- Big O Notations

- The magic you are looking for is in the work you are avoiding.

AlgoPrep

Quiz

↳ How many numbers are in range [3,10] {Corners included}

{3 4 5 6 7 8 9 10} → 8

inc, inc [a, b] → b-a+1	inc, exc [a, b) → b-a	exc, exc (a, b) → b-a-1

//log basics

$x * 2 = 10 \Rightarrow x = 10/2$

$x + 2 = 10 \Rightarrow x = 10 - 2$

$x^3 = 27 \Rightarrow x = \sqrt[3]{27}$

$x^2 = 16 \Rightarrow x = \sqrt{16}$

$x^2 = 16 \rightarrow 2 = \log_x 16$

ans $= \log_2 10$

AlgoPrep

→ $\log_a b$

number

representation

base of log

→ $\log_2 10 = ans$

$10 = 2^{ans} \rightarrow 2^{3.33}$

$\Downarrow$

ans = 3.33 . . .

(a & b are given)

* $\log_b a = ans \Rightarrow a = b^{ans}$ → then hit & trial for ans.

(i) $\log_2 64 = ans$

$64 = 2^{ans} \rightarrow 2^6 = 64 \Rightarrow ans = 6$

(ii) $\log_3 343 = ans$

$343 = 3^{ans} \Rightarrow ans = 5.2 \ldots$

(iii) $\log_2 33 = ans$

$33 = 2^{ans} \Rightarrow ans = 5.$ something

AlgoPrep

Properties:

↳ (I) $\log_a a^n = n$

↳ (II) $\log_c (a*b) = \log_c a + \log_c b$

ex: $\log_3 \underset{(5*2)}{10} = \log_3 5 + \log_3 2$

AlgoPrep

Quiz

↳ No. of steps for $N \rightarrow \frac{N}{2} \rightarrow \frac{N}{4} \rightarrow \cdots\cdots \rightarrow 1$

$$\left(\left(N * \frac{1}{2}\right) * \frac{1}{2}\right) * \frac{1}{2} \cdots\cdots = 1$$

No. of steps == How many times you can multiply by $\frac{1}{2}$ before reaching 1.

$$\frac{N}{2^K} = 1$$

↳ $N = 2^K \Rightarrow 2^K = N$

$K = \log_2 N$

AlgoPrep

A.P → Arithmetic Progression

$$3 \xrightarrow{+4} 7 \xrightarrow{+4} 11 \xrightarrow{+4} 15 \xrightarrow{+4} 19 \xrightarrow{+4} 23 \ldots$$

first term = a : Starting number, $a = 3$

Common diff = d : difference betn cons. no.s, $d = 4$

$$\overset{1}{(a+0*d)} \quad \overset{2}{(a+d)} \quad \overset{3}{(a+2d)} \quad \overset{4}{(a+3d)} \ldots \quad \overset{N\text{th term}}{a+(n-1)*d}$$

↳ Sum of first N terms of A.P $= \frac{N}{2}\left[2*a + (N-1)*d\right]$

G.P → geometric Progression

$$3 \xrightarrow{*2} 6 \xrightarrow{*2} 12 \xrightarrow{*2} 24 \xrightarrow{*2} 48 \ldots$$

$$50 \xrightarrow{*1/2} 25 \xrightarrow{*1/2} 12.5 \xrightarrow{*1/2} 6.25$$

first term = a : Starting number $a = 3$

Common ratio = r : multiple to get next no.

$$\overset{1}{ar^0} \quad \overset{2}{a*r} \quad \overset{3}{a*r^2} \quad \overset{4}{a*r^3} \ldots \quad \overset{N\text{th term}}{ar^{N-1}}$$

↳ Sum of first N terms of G.P $= a * \frac{r^N - 1}{r - 1}$

AlgoPrep

Quiz

```
int Sum = 0;
for (int i=1; i<=N; i++){
    Sum = Sum + 1;
}
```

$[1, N] \rightarrow N - \cancel{1} + \cancel{1}$

$\Downarrow$

N iterations

$\Downarrow$

$O(N)$

Quiz

```
void func (int N, int m){
    for (int i=1; i<=N; i++){
        Point (i);
    }
    for (int i=1; i<=M; i++){
        Point (i);
    }
}
```

$\rightarrow [1, N] = N - 1 + 1$

$+$

N iterations

$\rightarrow [1, M] = M - 1 + 1$

$+$

M iterations

Total = N+M iterations

$O(N+M) \xrightarrow{N>M} O(N)$

$O(N+M) \xrightarrow{N<M} O(M)$

AlgoPrep

Quiz

```
int fun (int N){
    int S=0;
    for (int i=0; i<=100; i++){
        S= S+ i^2;
    }
    return S;
}
```

→ [0, 100]

↳ 100 − 0 + 1

↳ 101 iterations

↓

O(1)

AlgoPrep

Quiz

```
void fun (int N){
    int S=0;
    for (int i=1; i*i<=N; i++){
        S= S+ i^2;
    }
    return S;
}
```

→ [1, √N] → √N − 1 + 1

= √N iterations

⇓

O(√N)

Quiz

```
void fun (int N){
    int i = N;
    while (i>=1){
        i = i/2;
    }
}
```

$N \rightarrow N/2 \rightarrow N/4 \rightarrow N/8 \rightarrow \cdots \rightarrow 1$

→ log N iterations

↓

O(log N)

AlgoPrep

Quiz

```
void fun (int N){

    int S=0;
    for (int i=0; i<=N; i=i*2){
        S= S+ i;
    }

}
```

→ [0, 0, 0, 0 - ———]

infinite

Quiz

```
void fun(int N){
    int S=0;
    for(int i=1; i<=N; i=i*2){
        S= S+ i;
    }
}
```

→ [1 → 2 → 4 → 8 . . — . . . . . — N]

↳ $\log N$ iterations

$\downarrow$

$O(\log N)$

$N \to \frac{N}{2} \to \frac{N}{4} \to \frac{N}{8} \to \dots \to 1$

$1 \to 2 \to 4 \to 8 \to \dots \to N$

Break till 9:25 pm

P1 → 1 2 3 4 5 + 5

P2 → 1 2 3 4 5 + 5

P3 → 1 2 3 4 5 + 5

P4 → 1 2 3 4 5 + 5

P5 → 1 2 3 4 5 + 5

→ 25

AlgoPrep

Nested loops

Quiz

```
void fun (int N){
    int S=0;
    for (int i=1; i<=10; i++){
        for (int j=1; j<=N; j++){
            S=S+10;
        }
    }
}
```

i	j	count
1	[1,N]	N
2	[1,N]	+ N
3	[1,N]	+ N
⋮		+
10	[1,N]	+ N
		10*N iteration

⇓

$O(N)$

Quiz

```
void fun (int N){
    int S=0;
    for (int i=1; i<=N; i++){
        for (int j=1; j<=N; j++){
            S=S+10;
        }
    }
}
```

i	j	count
1	[1,N]	N
2	[1,N]	+ N
3	[1,N]	+ N
⋮	⋮	+ ⋮
N	[1,N]	+ N
		(N*N) iterations

⇓

$O(N^2)$

Quiz

```
void fun (int N){
    int s=0;
    for (int i=1; i<=N ; i++){
        for (int j=1; j<=i ; j++){
            s=s+10;
        }
    }
}
```

i	j	count
1	[1,1]	1
2	[1,2]	2
3	[1,3]	3
⋮	⋮	⋮
N	[1,N]	N

$\frac{N*(N+1)}{2}$ iterations

↳ $O(N^2)$

Quiz

```
void fun (int N){
    for (int i=1; i<=2^N; i++){
        Print(i);
    }
}
```

→ $[1, 2^N]$ → 2^N iterations

⇓

$O(2^N)$

Quiz

```
void fun (int N){
    int s=0;
    for(int i=1; i<=N; i++){
        for(int j=1; j<=2^i; j++){
            s=s+10;
        }
    }
}
```

i	j	count
1	$[1, 2^1]$	2^1
2	$[1, 2^2]$	$+ 2^2$
3	$[1, 2^3]$	$+ 2^3$
⋮	⋮	⋮
N	$[1, 2^N]$	$+ 2^N$

$\Rightarrow$ sum of first N terms of G.P $= a * \frac{r^N - 1}{r - 1}$

$$\underbrace{2^1 + 2^2 + 2^3 + 2^4}_{*2 \quad *2 \quad *2} \ldots ____ + 2^N$$

$$= 2 * \frac{2^N - 1}{2 - 1} = 2 * (2^N - 1)$$

iterations

$\Downarrow$

$O(2^N)$

AlgoPrep

Comparison of iteration → $N = 10^5$

$$1 < \log_2 N < \sqrt{N} < N < N\log N < N\sqrt{N} < N^2 < 2^N$$

Time Complexity

↳ Big O Notation → Approx iteration count

(i) Calculate iteration.
(ii) around '+' sign, neglect lower order term.
(iii) Neglect Constants.

iteration count

ex: $\cancel{10}N^2 + \cancel{20N} + \cancel{30N^0}$

$\Downarrow$

$O(N^2)$

ex: $\cancel{6}N^2 + 15N\cancel{\log N} + \cancel{20N}$

$\Downarrow$

$O(N^2)$

AlgoPrep

ex: $\cancel{10}\, N\log N + \cancel{15N} + \cancel{60}$

$\Downarrow$

$O(N\log N)$

↳ No. of iterations for same Code is different

$N = 20$, $K = 100$

```
for (int i=1; i<=N; i++){
    if (i==K){
        break;
    }
}
```

→ 2 iterations

→ 10 iterations

↳ N iterations

Big O notation will Consider the worst case Scenario.

↳ $O(N)$

AlgoPrep